

Homework 4 – Biological Oceanography

Total 10 points

Due date: Tuesday 07/26/2022

1- Choose one of the animals from the SIO Vertebrate Collection and answer (3 Pts)

- a. What is the common and scientific names of the chosen species?
- b. List its taxonomic classification and what are the characteristics of each group.
- c. What type of marine community does this organism belong to and describe its geographic.
- d. What is the ecological role of this animal in the ecosystems it belongs to?

2- Choose one of the animals from the SIO Invertebrate Collection and answer (3 Pts)

- a. What is the common and scientific names of the chosen species?
- b. List its taxonomic classification and what are the characteristics of each group.
- c. What type of marine community does this organism belong to and describe its geographic.
- d. What is the ecological role of this animal in the ecosystems it belongs to?

3- Explain as detailed as possible how energy is captured (enters), stored, and transported through biological communities. (2 pts)

4- Read the article from Clark (2018)

<https://www.whoi.edu/oceanus/feature/the-recipe-for-a-harmful-algal-bloom/> and answer (2 pts)

- a. Explain why predicting *Pseudo-nitzschia* blooms are important and extremely challenging?
- b. How scientists use complex and numerous environmental observations to predict HABs?

- c. If *Pseudo-nitzschia* blooms were only a function of the ocean temperature, what do you predict will happen to the frequency and intensity of HABs as a consequence of global warming.

Mandatory Group Accountability Exercise: In a short paragraph or a statement, describe the role of each group member during the homework assignment.

ANY work done will be rewarded with a grade. Students with larger workloads will be rewarded extra points. Students with **NO CONTRIBUTION** to the group activity won't receive any points.

Example 1: All group members equally contributed to the homework assignment.

Example 2: Jane Doe answered questions 1 and 2, while John Doe, Jim Doe, and Carry Doe equally contributed to questions 3, 4, and 5.

Example 3: Everyone contributed except John Doe.

(In example 2 Jane Doe will receive extra 0.5 point. In example 3, John Doe will fail the assignment.)